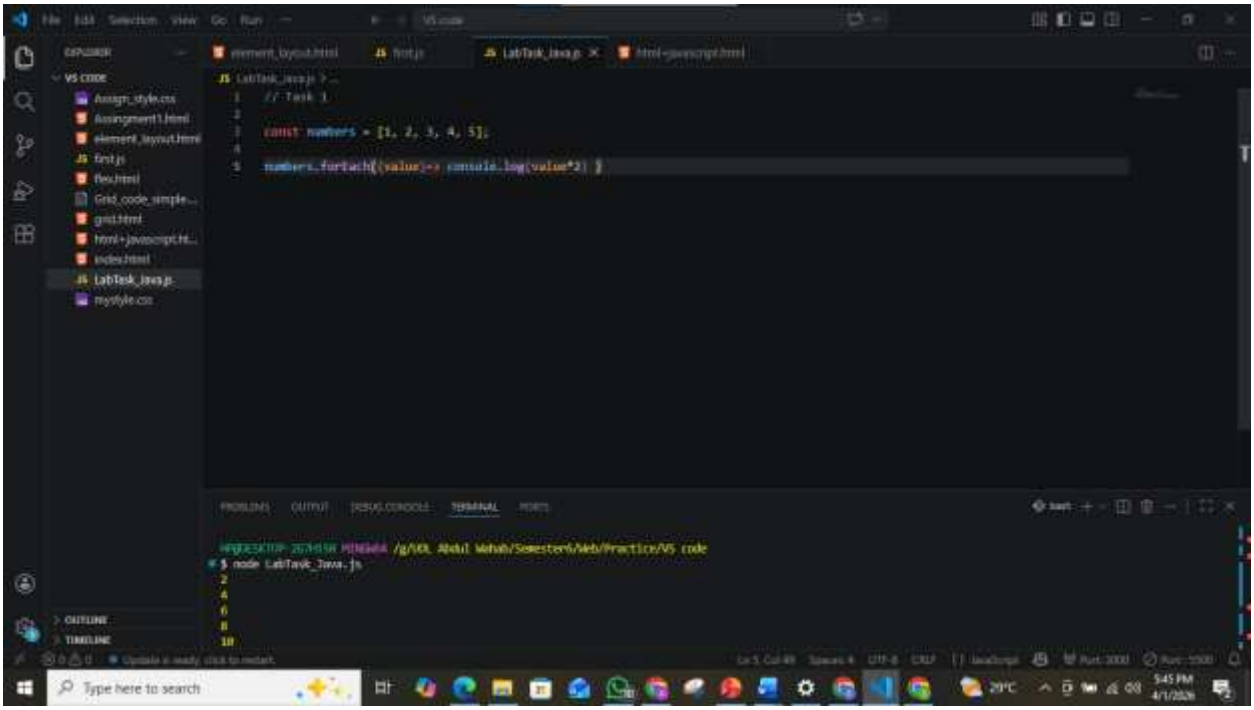
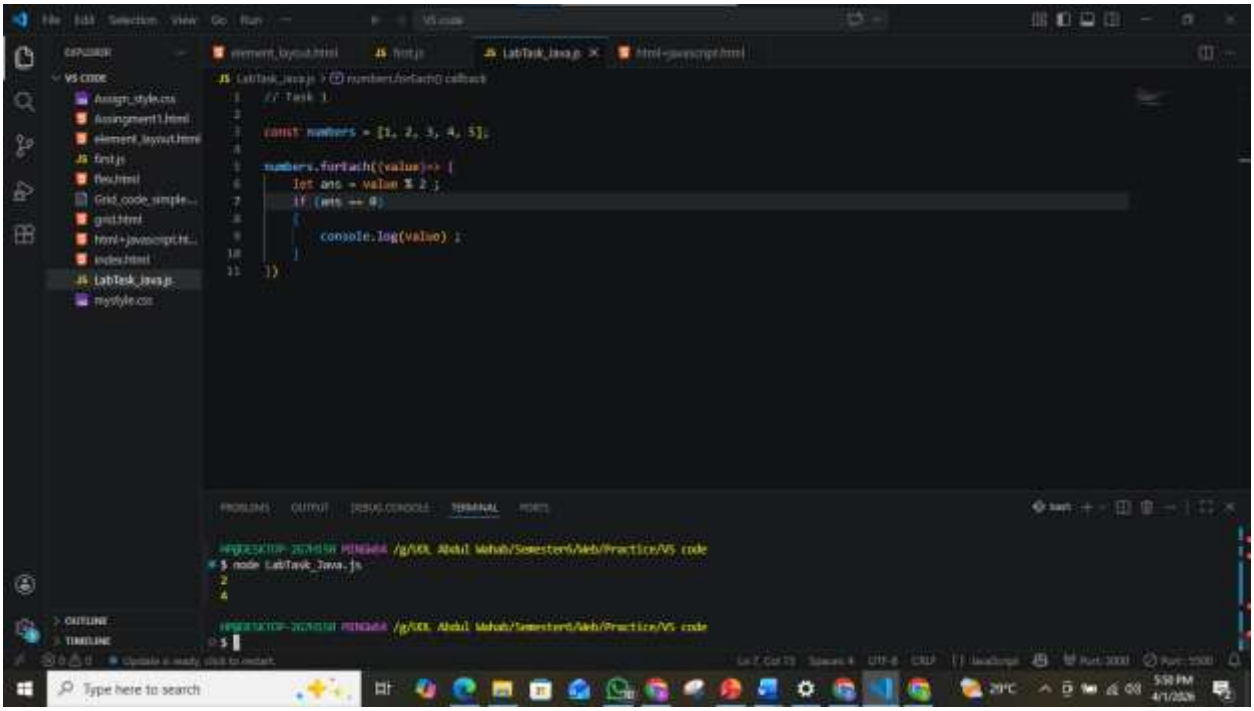
Name : Sheikh Abdul Wahab

ID : 70146066

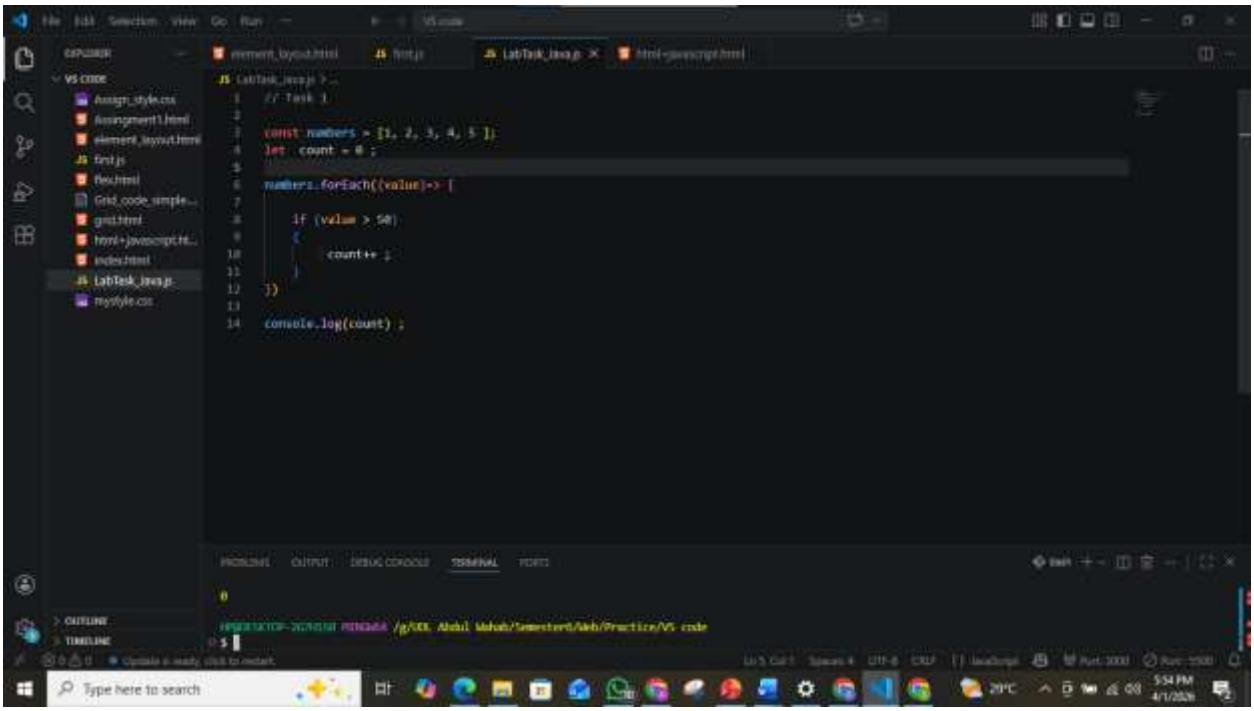
Task :1



Task: 2


// Task 1
const numbers = [1, 2, 3, 4, 5];
numbers.forEach((value)=> {
let ans = value % 2 ;
if (ans == 0)
{
console.log(value) ;
}
})
$ node LabTask_Java.js
2
4

Task: 3


// Task 1
const numbers = [1, 2, 3, 4, 5];
let count = 0 ;
numbers.forEach((value)=> {
if (value > 50)
{
count++ ;
}
})
console.log(count) ;
0

Task :4

```
numbers.forEach((value)=> {
    }
})

console.log(count) ;

//TAsk 4

const students = [
    {name : "Ali" , marks : 80},
    {name : "Sara" , marks : 45},
    {name : "John" , marks : 60}
]

students.forEach((value)=> {
    if(value.marks >= 50)
        console.log(value.name) ;


})
```

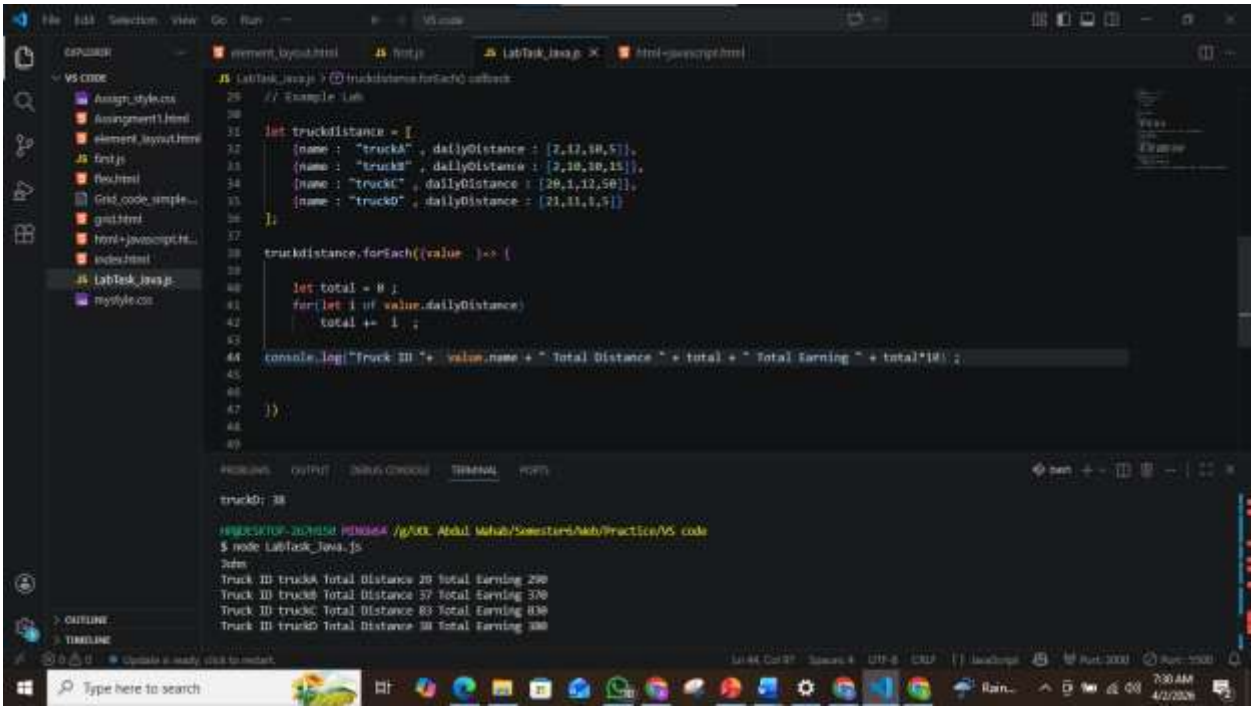
```
$ node LabTask_Java.js
0
Ali
John
Truck ID truckA Total Distance 29 Total Earning 290
Truck ID truckB Total Distance 37 Total Earning 370
Truck ID truckC Total Distance 83 Total Earning 830
Truck ID truckD Total Distance 38 Total Earning 380
```

Lab Example

```
// Example Lab

let truckdistance = [
    {name :  "truckA" , dailyDistance : [2,12,10,5]},
    {name :  "truckB" , dailyDistance : [2,10,10,15]},
    {name : "truckC" , dailyDistance : [20,1,12,50]},
    {name : "truckD" , dailyDistance : [21,11,1,5]}
]

truckdistance.forEach((value  )=> {

    let total = 0 ;
    for(let i of value.dailyDistance)
        total +=  i  ;

console.log(value.name + ": " + total) ;


})
```

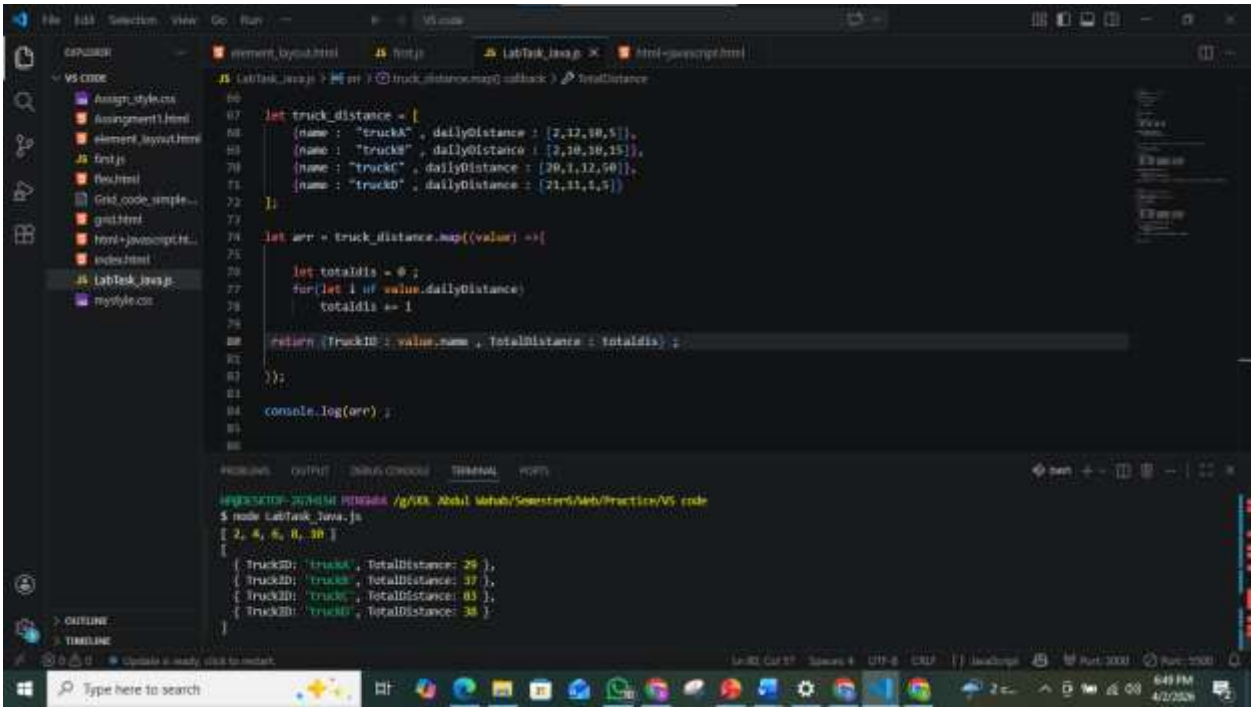
```
truckD 38

$ node LabTask_Java.js
John
truckA: 29
truckB: 37
truckC: 83
truckD: 38
```

Task : 5

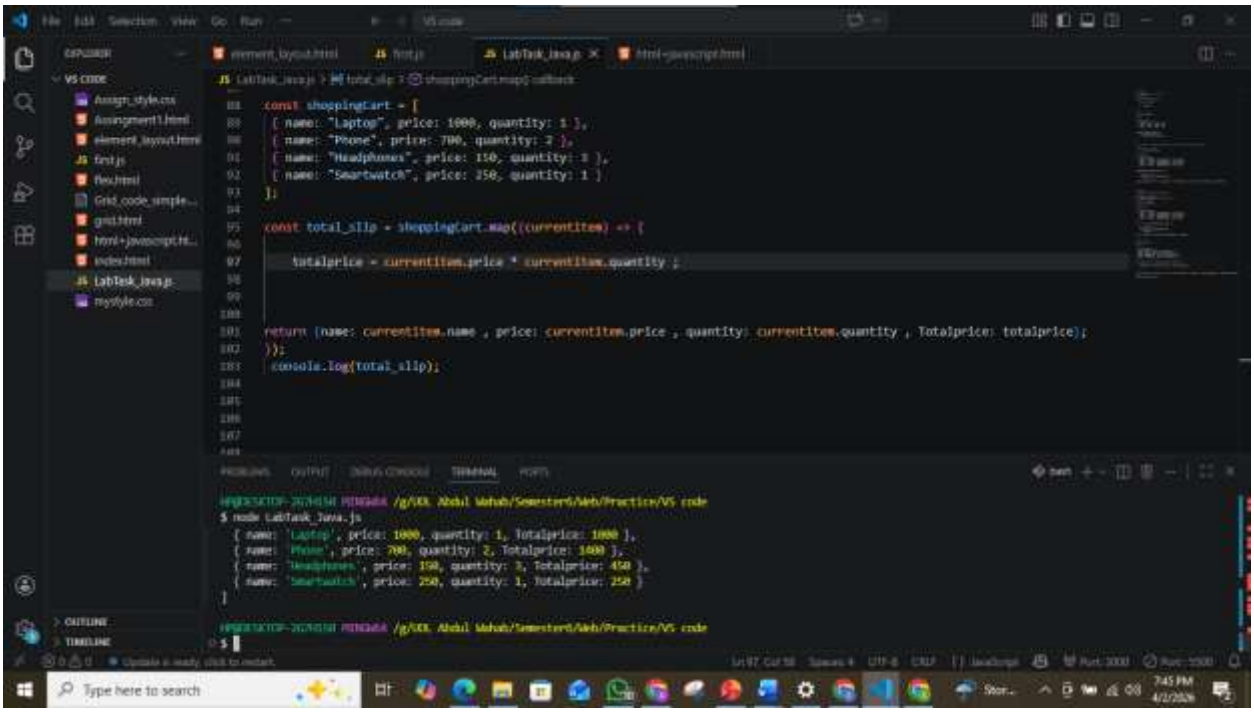


Map()

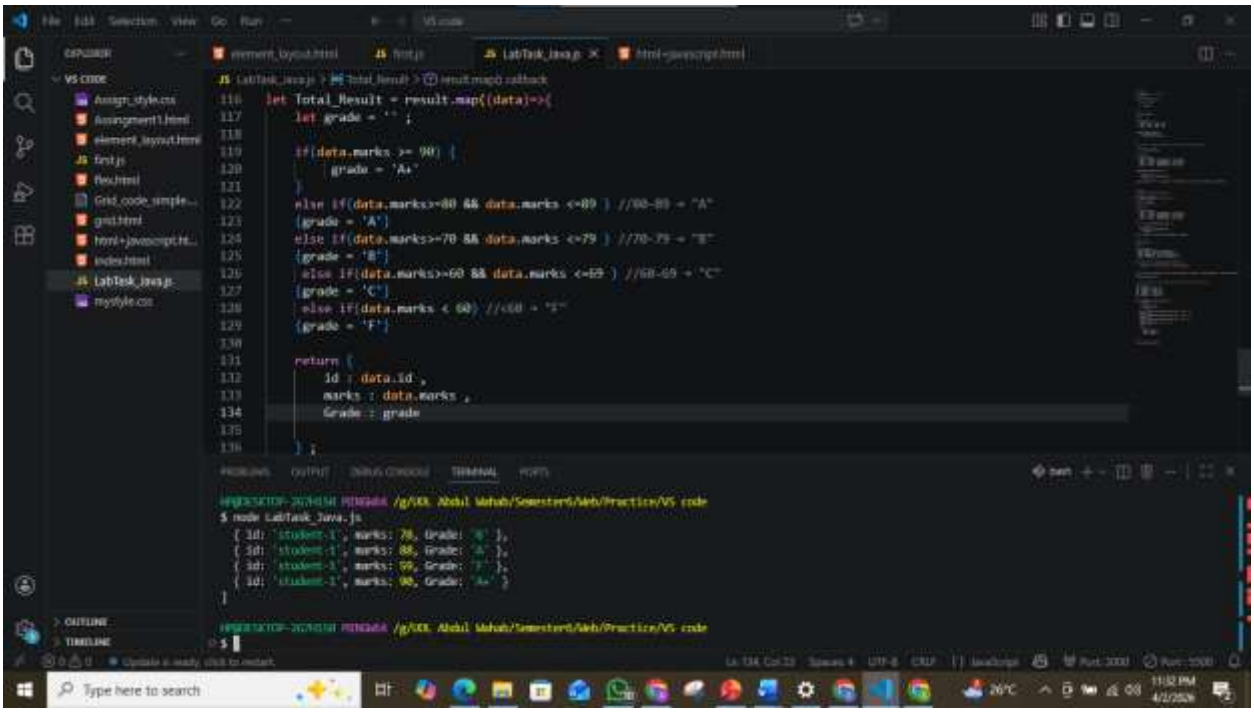
Example : 2


let truck_distance = [
{name : "truckA" , dailyDistance : [2,12,10,5]},
{name : "truckB" , dailyDistance : [2,10,10,15]},
{name : "truckC" , dailyDistance : [20,1,12,50]},
{name : "truckD" , dailyDistance : [21,11,1,5]}
];
let arr = truck_distance.map((value) =>{
let totaldis = 0 ;
for(let i of value.dailyDistance)
totaldis += i
return {TruckID : value.name , TotalDistance : totaldis} ;
});
console.log(arr) ;
$ node LabTask_Java.js
[2, 4, 6, 8, 10]
[
{ TruckID: 'truckA', TotalDistance: 29 },
{ TruckID: 'truckB', TotalDistance: 37 },
{ TruckID: 'truckC', TotalDistance: 83 },
{ TruckID: 'truckD', TotalDistance: 38 }
]

Task :6

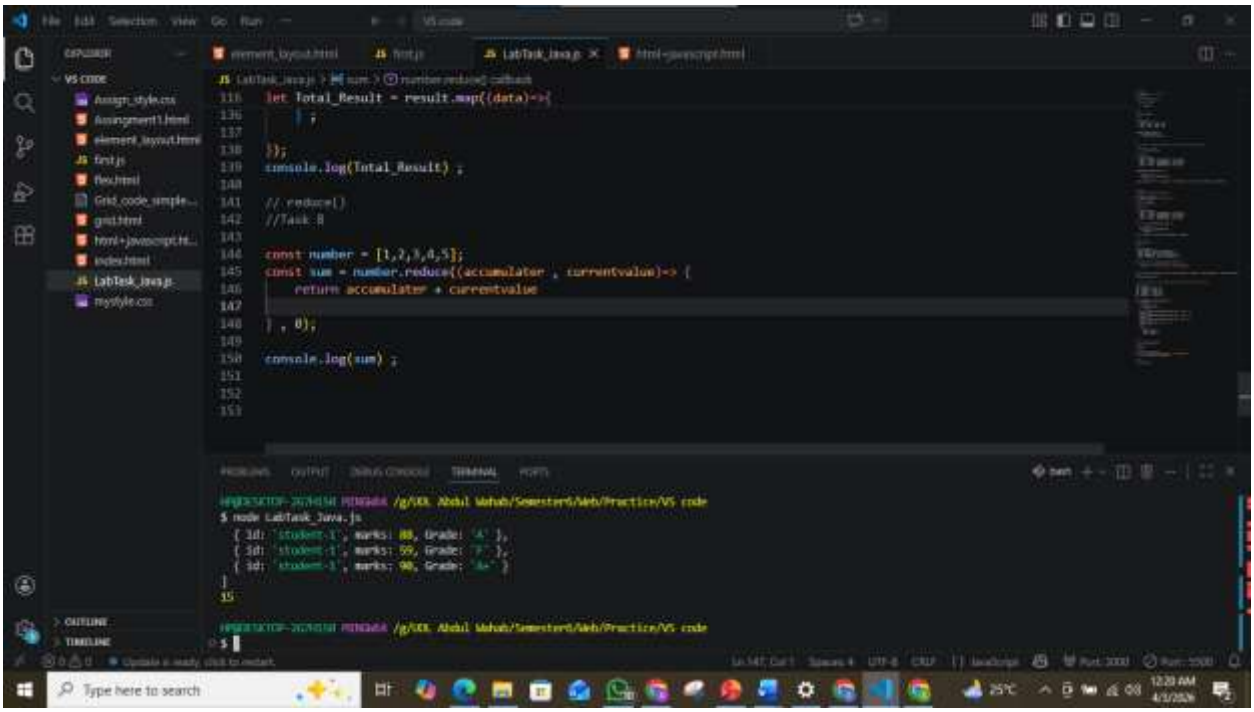

const shoppingCart = [
{ name: "Laptop", price: 1000, quantity: 1 },
{ name: "Phone", price: 700, quantity: 2 },
{ name: "Headphones", price: 150, quantity: 3 },
{ name: "Smartwatch", price: 250, quantity: 1 }
];
const total_slip = shoppingCart.map((currentitem) => {
totalprice = currentitem.price * currentitem.quantity ;
return {name: currentitem.name , price: currentitem.price , quantity: currentitem.quantity , Totalprice: totalprice};
});
console.log(total_slip);
$ node LabTask_Java.js
{ name: 'Laptop', price: 1000, quantity: 1, Totalprice: 1000 },
{ name: 'Phone', price: 700, quantity: 2, Totalprice: 1400 },
{ name: 'Headphones', price: 150, quantity: 3, Totalprice: 450 },
{ name: 'Smartwatch', price: 250, quantity: 1, Totalprice: 250 }
]

Task : 7

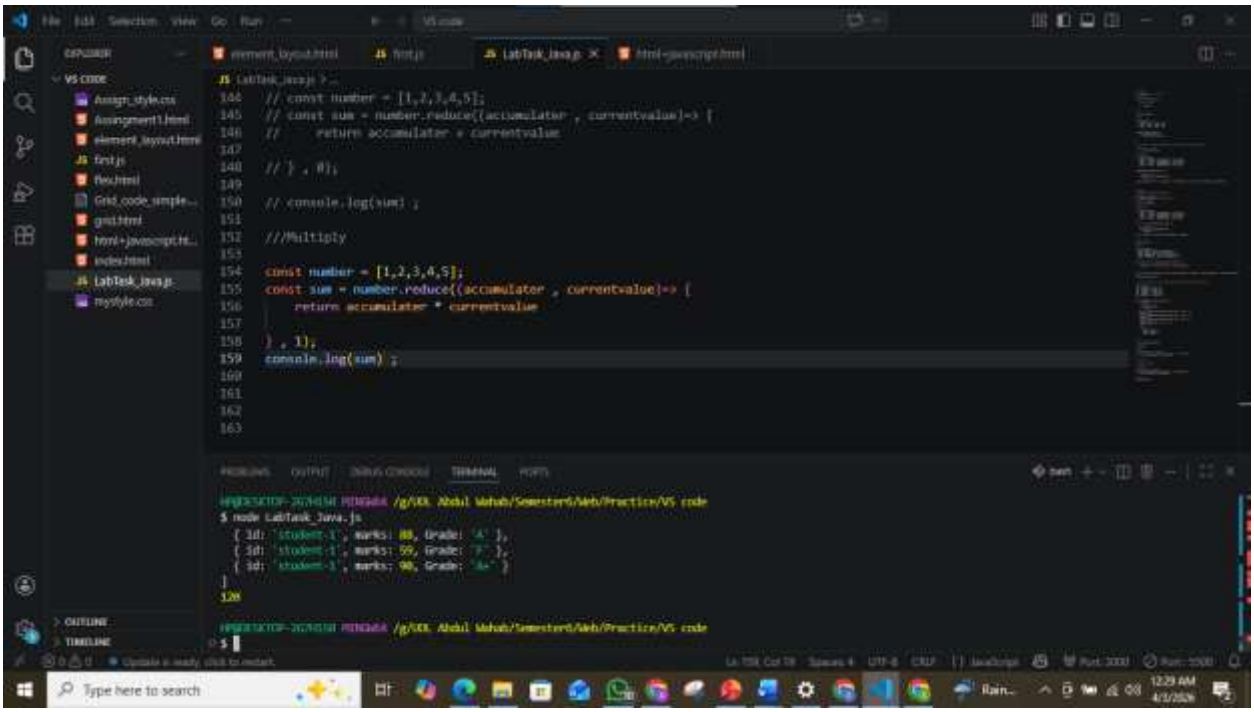


reduce()

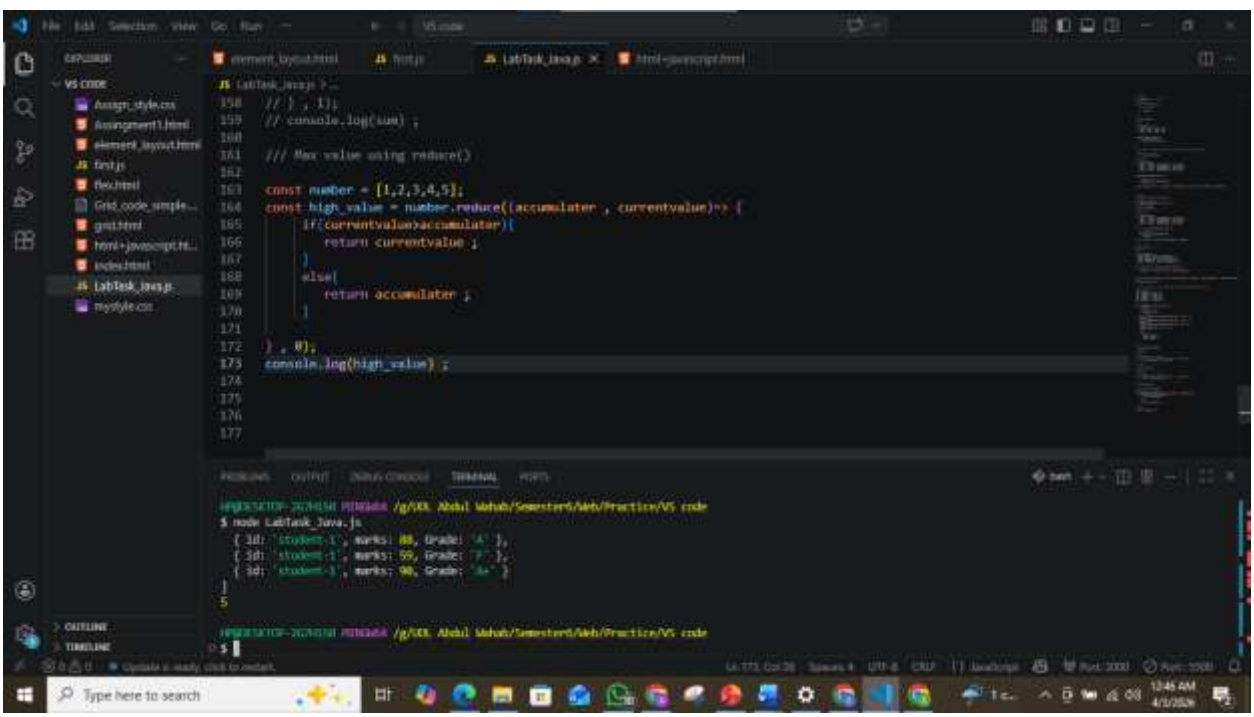
task:8



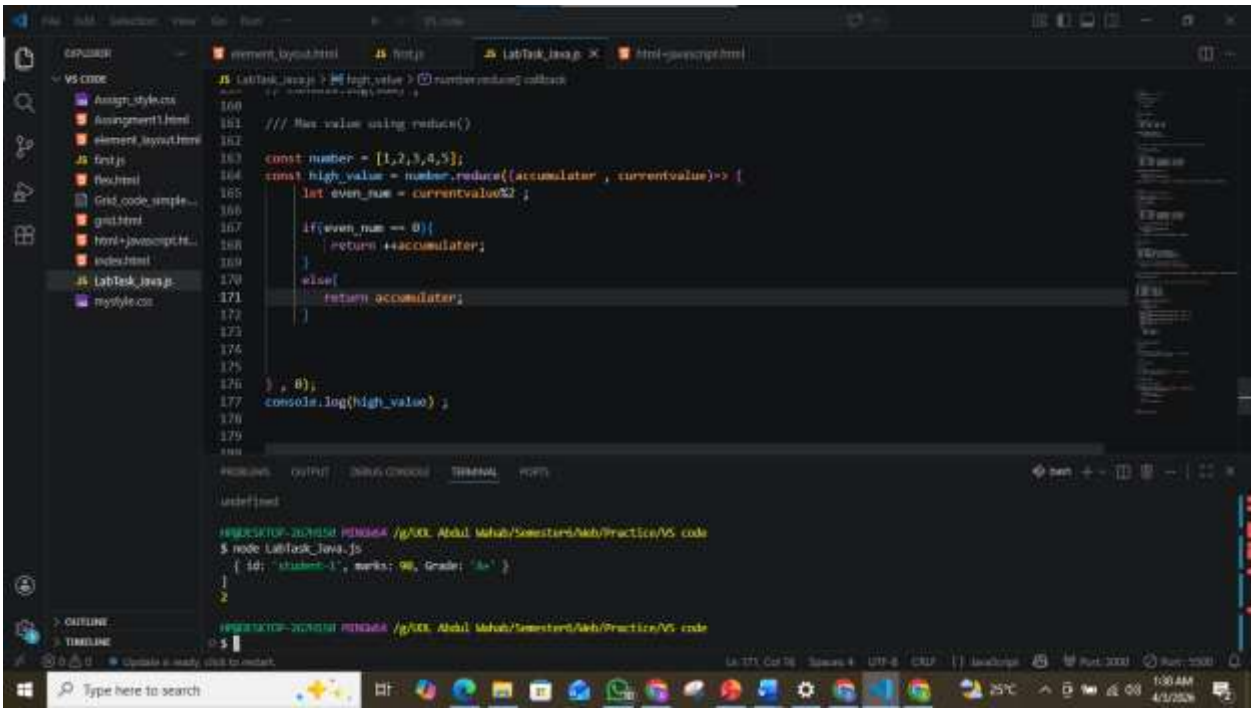
ii) multiply



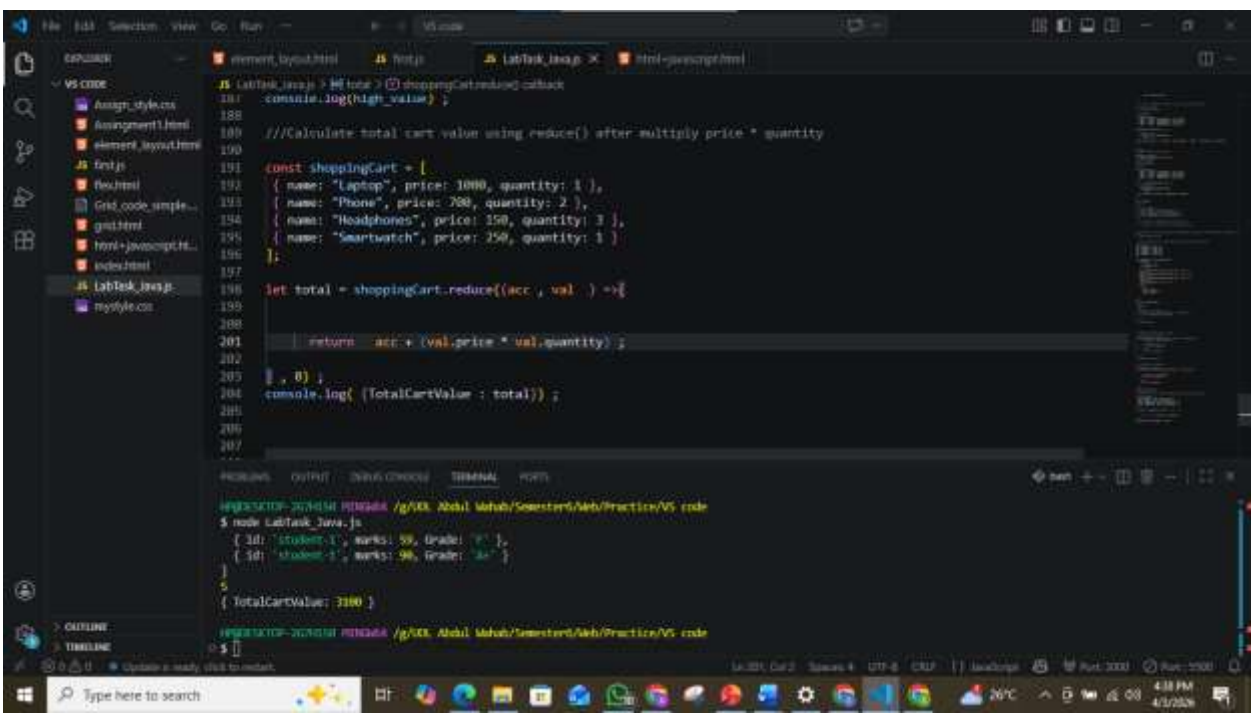
iii) Max value find



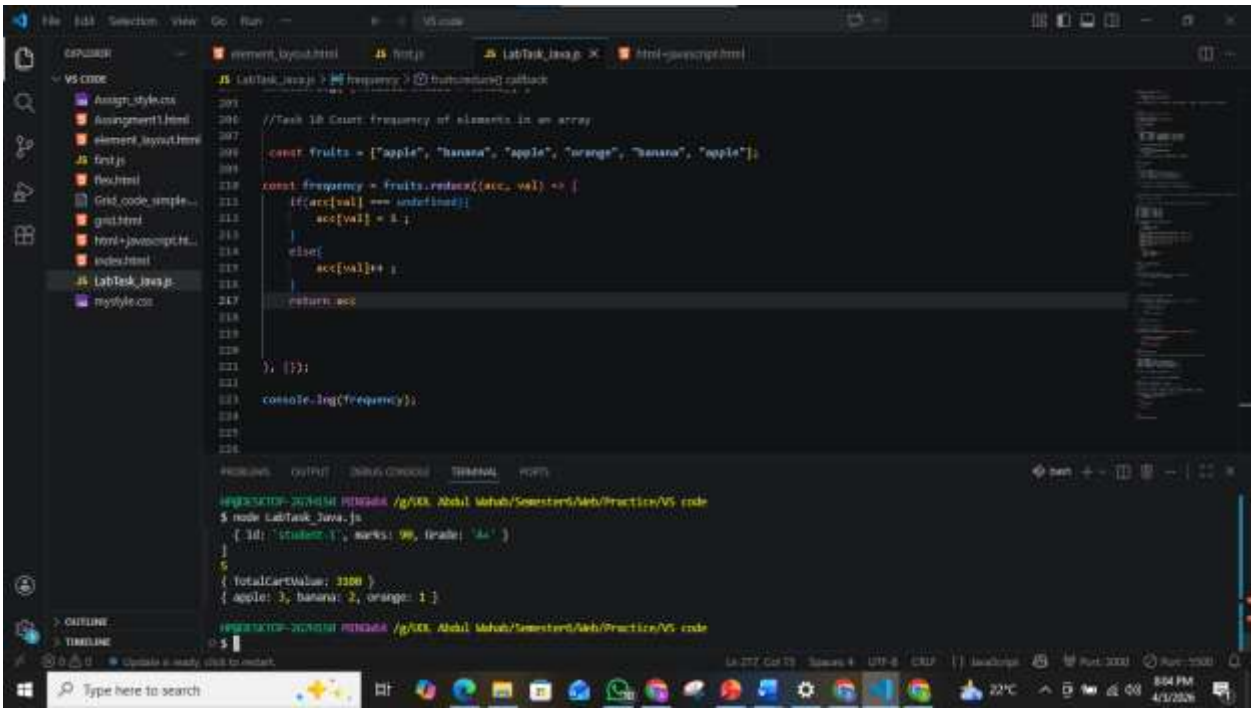
iv) count how many even numbers using reduce()



Task 9

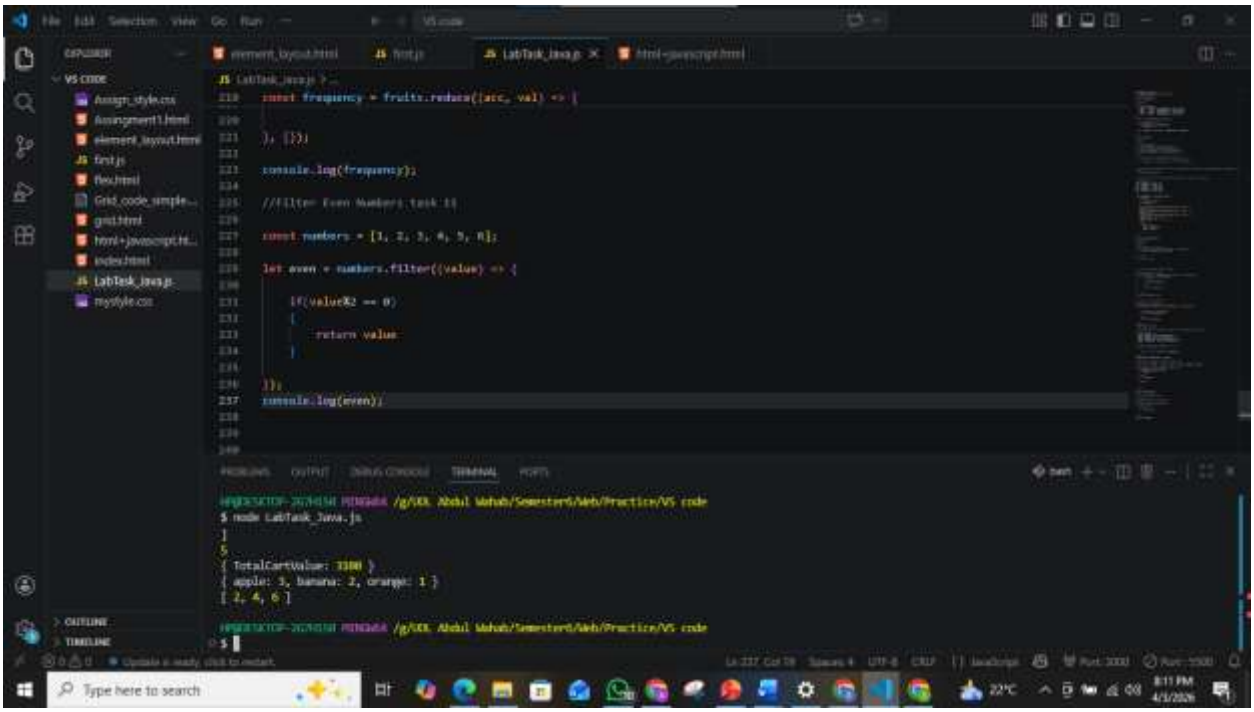


Task 10

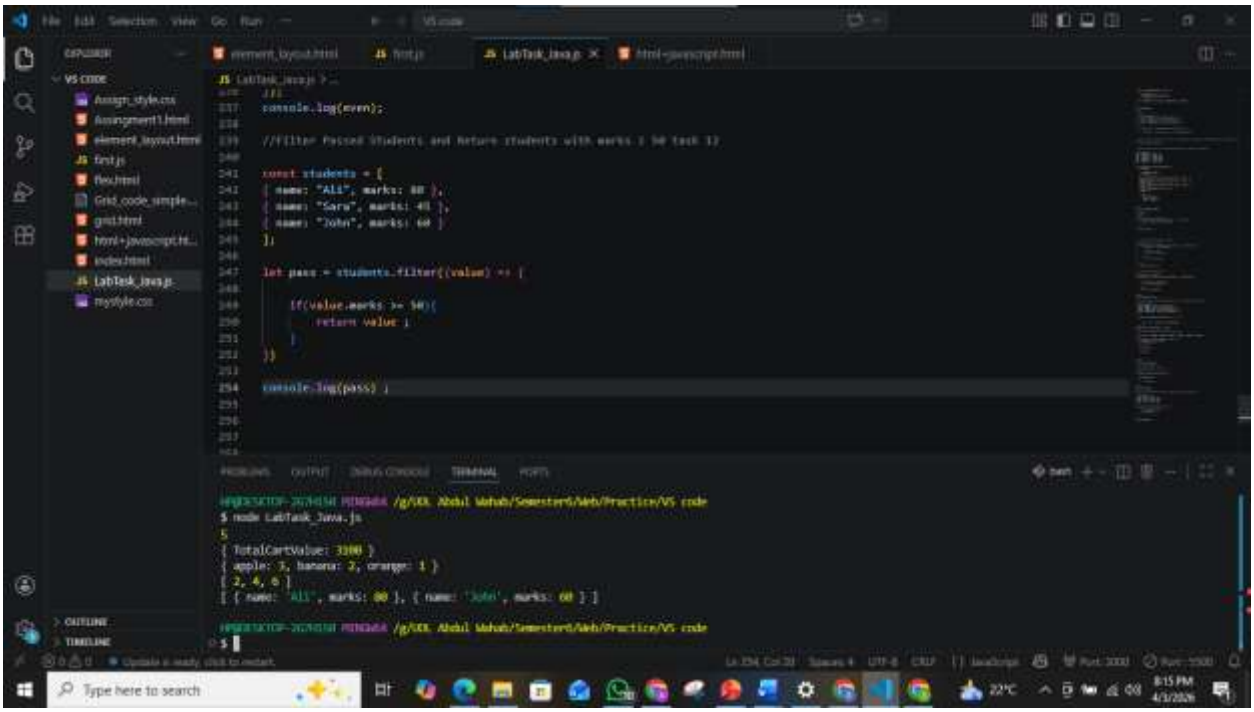


Filter ()

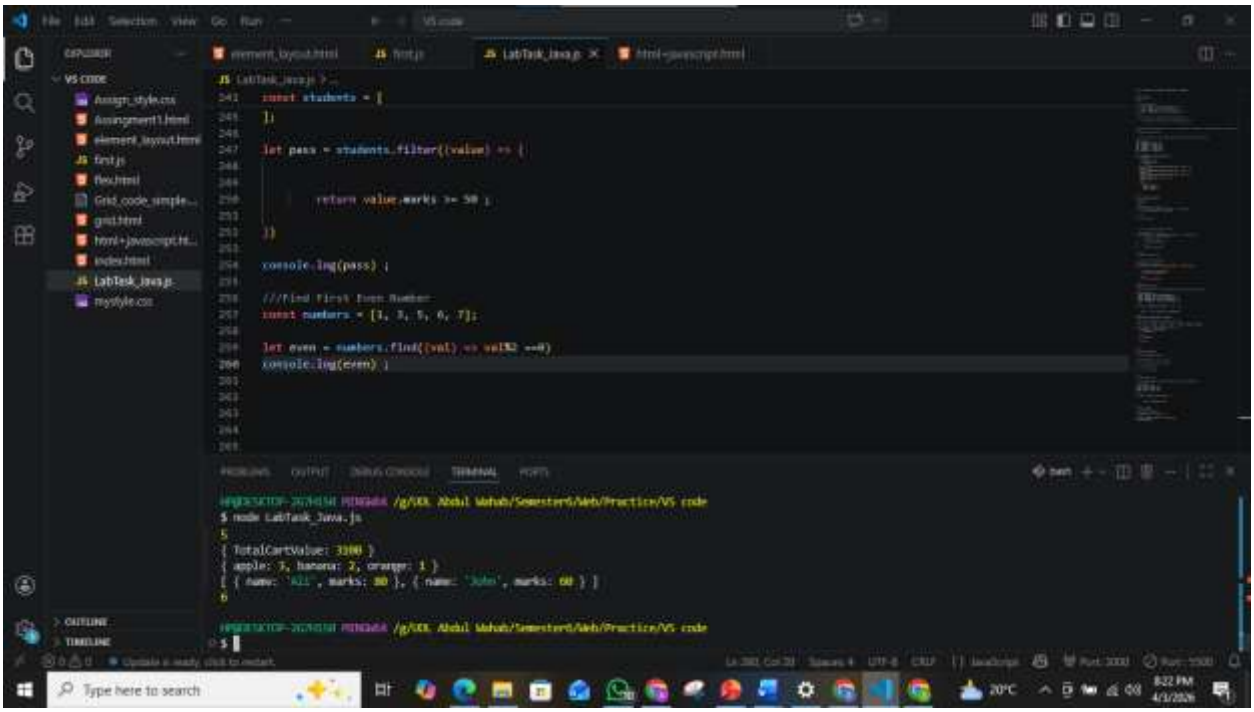
Task 11



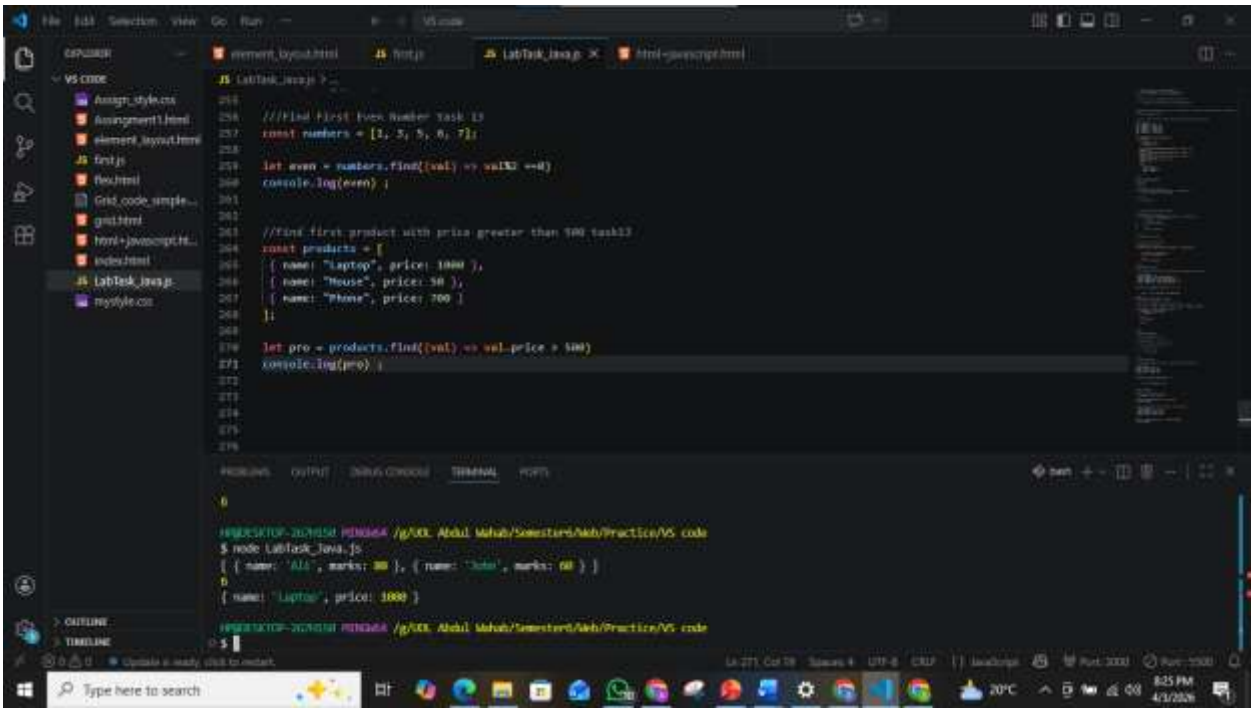
Task 12



Task 13



Task 13



// Task 1

// const numbers = [1, 2, 3, 4, 5];

// let count = 0 ;

// numbers.forEach((value)=> {

//   if (value > 50)

//   {

//   count++ ;

//   }

// })

```
// console.log(count) ;

//TAsk 4

// const students = [
//    {name : "Ali" , marks : 80},
//    {name : "Sara" , marks : 45},
//    {name : "John" , marks : 60}
// ]

// students.forEach((value)=> {
//    if(value.marks >= 50)
//       console.log(value.name) ;



// })

// let result = students.filter((value , index , array )=> value.marks>=50) ;

// for(let i of result)
//    console.log(i.name);

// Example Lab

let truckdistance = [
   {name :  "truckA" , dailyDistance : [2,12,10,5]},
```

```
  {name :  "truckB" , dailyDistance : [2,10,10,15]},
  {name : "truckC" , dailyDistance : [20,1,12,50]},
  {name : "truckD" , dailyDistance : [21,11,1,5]}
];

truckdistance.forEach((value  )=> {

  let total = 0 ;
  for(let i of value.dailyDistance)
     total +=  i  ;

console.log("Truck ID "+  value.name + " Total Distance " + total + " Total Earning " +
total*10) ;

})

/// map()
//Example 1: Multiply each number by 2

let num = [1,2,3,4,5];
let newarray = num.map(function(value){
  return value*2 ;
});
console.log(newarray)

//Example 2
```

```
let truck_distance = [
   {name :  "truckA" , dailyDistance : [2,12,10,5]},
   {name :  "truckB" , dailyDistance : [2,10,10,15]},
   {name : "truckC" , dailyDistance : [20,1,12,50]},
   {name : "truckD" , dailyDistance : [21,11,1,5]}
];

let arr = truck_distance.map((value) =>{

   let totaldis = 0 ;
   for(let i of value.dailyDistance)
      totaldis += i

 return {TruckID : value.name , TotalDistance : totaldis} ;

});

console.log(arr) ;

//Task 6

// const shoppingCart = [
//  { name: "Laptop", price: 1000, quantity: 1 },
//  { name: "Phone", price: 700, quantity: 2 },
//  { name: "Headphones", price: 150, quantity: 3 },
```

```
//  { name: "Smartwatch", price: 250, quantity: 1 }
// ];

// const total_slip = shoppingCart.map((currentitem) => {

//    totalprice = currentitem.price * currentitem.quantity ;



// return {name: currentitem.name , price: currentitem.price , quantity:
currentitem.quantity , Totalprice: totalprice};
// });
//  console.log(total_slip);

 ///Task 7 Create an array of student marks and return a new array with grades:

 const result = [
{id: 'student-1', marks: 45},
{id: 'student-1', marks: 78},
{id: 'student-1' , marks: 88},
{id: 'student-1' , marks: 59},
{id: 'student-1' , marks: 90}
];

let Total_Result = result.map((data)=>{
   let grade = '' ;
```

```
    if(data.marks >= 90) {
        grade = 'A+'
    }
    else if(data.marks>=80 && data.marks <=89 ) //80–89 → "A"
    {grade = 'A'}
    else if(data.marks>=70 && data.marks <=79 ) //70–79 → "B"
    {grade = 'B'}
     else if(data.marks>=60 && data.marks <=69 ) //60–69 → "C"
    {grade = 'C'}
     else if(data.marks < 60) //<60 → "F"
    {grade = 'F'}

    return {
       id : data.id ,
       marks : data.marks ,
       Grade : grade

    } ;

});
console.log(Total_Result) ;

// reduce()
//Task 8
```

```
// const number = [1,2,3,4,5];
// const sum = number.reduce((accumulater , currentvalue)=> {
//     return accumulater + currentvalue

// } , 0);

// console.log(sum) ;
```

```
/// count how many even numbers  reduce()

// const number = [1,2,3,4,5];
// const high_value = number.reduce((accumulater , currentvalue)=> {
//     let even_num = currentvalue%2 ;

//     if(even_num == 0){
//        return ++accumulater;
//     }
//     else{
//       return accumulater;
//     }

// } , 0);
// console.log(high_value) ;
```

```
/// find max value using reduce

const number = [1,2,3,4,5];
const high_value = number.reduce((accumulater , currentvalue)=> {



    if(currentvalue > accumulater){
       return  currentvalue;
    }
    else{
      return accumulater;
    }

} , 0);
console.log(high_value) ;

///Calculate total cart value using reduce() after multiply price * quantity

const shoppingCart = [
 { name: "Laptop", price: 1000, quantity: 1 },
 { name: "Phone", price: 700, quantity: 2 },
 { name: "Headphones", price: 150, quantity: 3 },
 { name: "Smartwatch", price: 250, quantity: 1 }
];
```

```
let total = shoppingCart.reduce((acc , val  ) =>{



     return   acc + (val.price * val.quantity) ;


} , 0) ;
console.log( {TotalCartValue : total}) ;


//Task 10 Count frequency of elements in an array


 const fruits = ["apple", "banana", "apple", "orange", "banana", "apple"];


const frequency = fruits.reduce((acc, val) => {
   if(acc[val] === undefined){
      acc[val] = 1 ;
   }
   else{
      acc[val]++ ;
   }
   return acc




}, {});


console.log(frequency);
```

```
//Filter Even Numbers task 11

// const numbers = [1, 2, 3, 4, 5, 6];

// let even = numbers.filter((value) => {

//    if(value%2 == 0)
//    {
//       return value
//    }

// });
// console.log(even);

//Filter Passed Students and Return students with marks ≥ 50 task 12

const students = [
{ name: "Ali", marks: 80 },
{ name: "Sara", marks: 45 },
{ name: "John", marks: 60 }
];

let pass = students.filter((value) => {
```

```
    return value.marks >= 50 ;

})

console.log(pass) ;

///Find First Even Number task 13
const numbers = [1, 3, 5, 6, 7];

let even = numbers.find((val) => val%2 ==0)
console.log(even) ;

//Find first product with price greater than 500 task13
const products = [
 { name: "Laptop", price: 1000 },
 { name: "Mouse", price: 50 },
 { name: "Phone", price: 700 }
];

let pro = products.find((val) => val.price > 500)
console.log(pro) ;
```